

An aerial photograph of Seattle, Washington, taken at sunset. The Space Needle is the central focus, with the sun setting directly behind its top, creating a bright glow. The city's skyline is visible in the foreground and middle ground, with various buildings and green spaces. The background shows the city extending to the water and distant mountains under a clear, orange-hued sky.

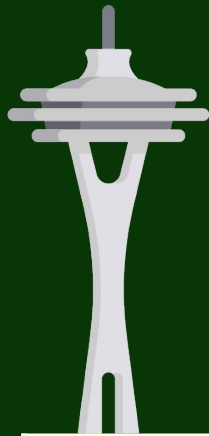
WA Landmark Classification

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Photo by Robert Ritchie on [Unsplash](#)

Project Background



The goal of our project is to create a website that can identify **landmarks in Washington state** (given an image).

Tool Project

- Python
- Streamlit

Image Classification

- TensorFlow
- Transfer Learning

Previous Work

- Landmark classification Kaggle challenge
- How ours is different?

Data Sources

Google Landmarks Dataset v2

- 4+ million images → 200,000 locations
- Narrowed to landmarks in WA state - reduced to 296 locations
- Fields: id, name, wikimedia image url

Wikimedia



- Location
- Supercategory
- Latitude and Longitude

Personal Photos

- Space Needle
- Suzzallo Allen Library
- Drumheller Fountain
- Historic Chinatown Gate

Use Case 1 - Classify an Image



User: the Tourist

Uploads an image to be classified



Model classifies image



Interface shows the result, the accuracy, the landmark category, and location to the user

Use Case 2 - Search for a Landmark



User: the Tourist



Searches the name of the landmark



Landmark classes are queried



Interface shows the user if the name is in the dataset and provides landmark details

Use Case 3 - Ask for a new landmark



Submits a feedback form with text
and/or images



Email sent to model maintainer



Model maintainer uses package
documentation to update model

User: the Model Maintainer

Component Design - Modular UI

Home.py

Landing page



Classifier.py

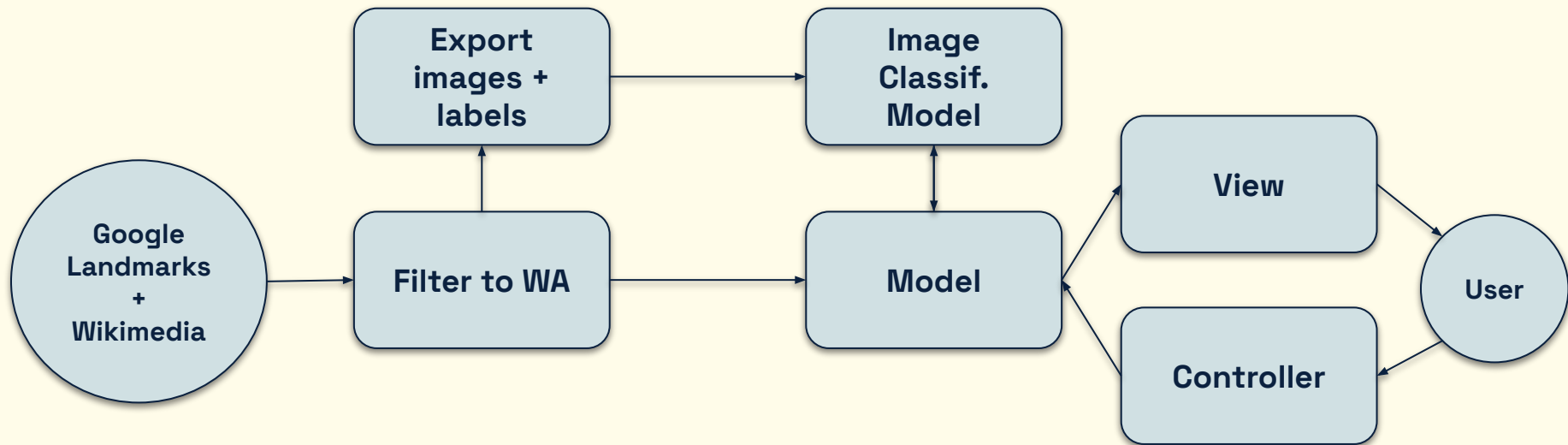
Search_page.py

Model_Analysis.py

Feedback.py

Component Design - Backend

Pipe & Filter + MVC



Live Demo!



Lessons Learned



Modularization of components

Organized code
Easy collaboration

Streamlit can be inflexible

Multipage app naming incompatible w pylint
AppTest limitations

Transfer learning rocks

Without transfer learning: ~30% accuracy
With transfer learning: 87% accuracy

PyTorch vs. TensorFlow

Switched to PyTorch after Technology Review
Final product uses TensorFlow

Future Work

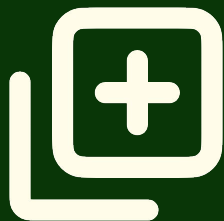
WA State GeoGuessr

- Fun way to engage users!



Expand set of Landmarks

- Requires manual data collection or merge of existing datasets



Location-based Landmark Filtering

- Query landmarks within user-specified location



A scenic landscape featuring a calm lake reflecting the surrounding forest and distant mountains. In the foreground, a wooden bench sits on a grassy bank. The scene is framed by tall evergreen trees on the left and a large tree on the right. A semi-transparent green shape with rounded corners is overlaid on the right side of the image, containing the text "Thank You!" in white.

Thank You!